



**AI-DRIVEN INTELLIGENT VIVA-VOCE  
EVALUATION SYSTEM USING NLP**



**A PROJECT REPORT**

*Submitted by*

|                                |                     |
|--------------------------------|---------------------|
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*in partial fulfillment of the requirements for the award degree of  
Bachelor of Engineering*

**20CS8504 PROJECT WORK**

**DEPARTMENT OF COMPUTER SCIENCE AND  
ENGINEERING**

**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY  
(AUTONOMOUS)**

**SAMAYAPURAM - 621112**

**APRIL 2026**



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**BONAFIDE CERTIFICATE**

The work embodied in the present project report entitled **AI-DRIVEN INTELLIGENT VIVA-VOCE EVALUATION SYSTEM USING NLP** has been carried out by the students **DENNIS CYRUS J, HARRISH RAGHAVENDAAR RR, JEEVAN T** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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## ABSTRACT

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) is designed to automate and improve the traditional viva examination process in educational institutions. Viva-voce examinations are commonly used to evaluate a student's conceptual understanding, communication ability, and subject knowledge. However, traditional viva examinations are often time-consuming and may involve subjective evaluation by examiners, which can lead to inconsistencies in grading. To address these challenges, the proposed system integrates Artificial Intelligence and Natural Language Processing techniques to analyze student responses and generate automated evaluation results. The system provides an interactive platform where students can respond to viva questions either through text or voice input. The responses are processed using NLP techniques such as tokenization, stop-word removal, semantic similarity analysis, and contextual understanding to evaluate the correctness and relevance of the answer.

**Keywords:** Artificial Intelligence, Natural Language Processing (NLP), Viva-Voce Evaluation, Automated Assessment, Machine Learning, Educational Technology, Semantic Analysis, Student Response Evaluation, Intelligent Examination System

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**SIGNATURE**

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## LIST OF ABBREVIATIONS

|      |   |                              |
|------|---|------------------------------|
| AI   | - | ARTIFICIAL INTELLIGENCE      |
| NLP  | - | NATURAL LANGUAGE PROCESSING  |
| ML   | - | MACHINE LEARNING             |
| ANN  | - | ARTIFICIAL NEURAL NETWORK    |
| NLG  | - | NATURAL LANGUAGE GENERATION  |
| UI   | - | USER INTERFACE               |
| UX   | - | USER EXPERIENCE              |
| ASR  | - | AUTOMATIC SPEECH RECOGNITION |
| STT  | - | SPEECH TO TEXT               |
| TTS  | - | TEXT TO SPEECH               |
| LMS  | - | LEARNING MANAGEMENT SYSTEM   |
| DB   | - | DATABASE                     |
| SQL  | - | STRUCTURED QUERY LANGUAGE    |
| HTML | - | HYPER TEXT MARKUP LANGUAGE   |
| CSS  | - | CASCADING STYLE SHEETS       |
| JS   | - | JAVASCRIPT                   |
| NLTK | - | NATURAL LANGUAGE TOOLKIT     |
| POS  | - | PART OF SPEECH               |

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 DESCRIPTION**

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) is designed to modernize the traditional oral examination process by integrating artificial intelligence and language processing technologies. Viva-voce examinations play an important role in evaluating a student's conceptual understanding, communication ability, and subject knowledge. However, traditional viva examinations often depend heavily on the examiner's judgment, which may sometimes introduce inconsistencies, bias, or time constraints when evaluating a large number of students. To address these challenges, intelligent automated systems can be developed to support and enhance the evaluation process.

Smart evaluation platform capable of conducting viva-voce sessions and analysing student responses using Natural Language Processing techniques. NLP enables machines to understand, interpret, and analyse human language in a meaningful way. By incorporating NLP, the system can process textual or speech-based responses provided by students and evaluate their relevance, accuracy, and completeness with respect to the question asked. This helps in creating a more structured and objective evaluation environment.

The system is designed to simulate an interactive viva examination environment where students are asked questions from predefined academic topics. The answers given by the students are analyzed through language processing techniques such as tokenization, semantic analysis, keyword extraction, and similarity matching. These processes allow the system to understand whether the student's answer reflects the expected concept or knowledge required for the question. Based on the analysis, the system generates evaluation scores and feedback automatically. Another important aspect of this system is its ability to handle large volumes of student evaluations efficiently.

## **1.2 ARTIFICIAL INTELLIGENCE AND NATURAL LANGUAGE PROCESSING**

The domain of this project lies primarily in the fields of Artificial Intelligence (AI) and Natural Language Processing (NLP). Artificial Intelligence focuses on developing systems capable of performing tasks that typically require human intelligence, such as reasoning, decision making, learning from data, and understanding language. Within AI, Natural Language Processing is a specialized area that enables computers to process and analyze human language in both written and spoken forms.

In the context of this project, NLP plays a central role in interpreting and evaluating student responses during viva-voce examinations. Traditional computer systems are designed to process structured data, whereas human language is highly unstructured and complex. NLP techniques allow computers to convert human language into a structured format that can be analyzed computationally. By using these techniques, the system can determine whether a student's answer contains relevant information and conceptual understanding related to the question.

Another important aspect of this domain is machine learning, which enables systems to improve their performance based on data and patterns. Machine learning models can be trained to recognize similarities between student responses and expected answers. This allows the system to evaluate answers even when students use different words or sentence structures to express the same concept. Such flexibility is essential for educational evaluation systems because students may present their knowledge in many different ways.

The AI and NLP domain also supports advanced capabilities such as speech recognition and semantic understanding. Speech recognition technologies can convert spoken responses into text, which can then be processed using NLP algorithms. Semantic analysis helps the system understand the meaning and context of words within a sentence rather than simply matching keywords. These technologies enable the system to evaluate responses in a more intelligent and meaningful manner.

### **1.2.1 Artificial Intelligence in Education**

Artificial Intelligence has become an important component in modern educational systems. AI technologies help automate complex tasks such as grading assignments, analyzing student performance, and providing personalized feedback. In the context of viva-voce evaluation, AI enables the development of intelligent systems that can simulate examiner-like behavior and analyze student responses effectively.

AI systems can process large datasets and identify patterns in student responses, which helps in creating more consistent evaluation criteria. By integrating AI into educational assessment systems, institutions can reduce manual effort while maintaining fairness and transparency in the evaluation process.

### **1.2.2 Natural Language Processing for Response Analysis**

Natural Language Processing is a branch of artificial intelligence that focuses on enabling computers to understand and process human language. In this project, NLP techniques are used to analyze student answers by breaking down sentences into smaller components and identifying key concepts.

Techniques such as tokenization, stop-word removal, stemming, and semantic similarity analysis help the system determine whether the student's response aligns with the expected answer. NLP also allows the system to interpret answers that may be phrased differently but convey the same meaning. This capability makes the evaluation process more flexible and accurate.

In addition, Natural Language Processing enables the system to evaluate responses not only based on keywords but also by understanding the context and intent of the student's answer. By applying techniques such as sentence parsing, part-of-speech tagging, and contextual similarity analysis, the system can determine whether the response logically relates to the question being asked.

### **1.2.3 Intelligent Assessment Systems**

Intelligent assessment systems use AI algorithms to automatically evaluate student responses and generate feedback. These systems are designed to replicate certain aspects of human evaluation while maintaining consistency and objectivity.

In the case of viva-voce evaluation, intelligent systems analyze the quality of student answers, determine their relevance to the question, and assign evaluation scores accordingly. Such systems help educational institutions manage assessments more efficiently and provide students with immediate feedback on their performance.

## **1.3 EDUCATIONAL TECHNOLOGY**

Educational Technology refers to the use of digital tools and technologies to improve teaching, learning, and assessment processes. Modern educational environments increasingly rely on technology-based platforms to support online learning, digital examinations, and automated evaluation systems.

The AI-Driven Viva-Voce Evaluation System falls within the Educational Technology domain because it enhances the traditional assessment process through automation and intelligent analysis. By integrating AI and NLP into educational evaluation, the system provides a scalable solution for conducting oral examinations in academic institutions.

## **1.4 LANGUAGE PROCESSING AND ANALYTICS**

Language processing and analytics involve analyzing textual or spoken data to extract meaningful insights. This domain is particularly relevant for applications that require understanding human communication, such as chatbots, virtual assistants, and automated evaluation systems.

In the proposed project, language processing techniques are used to analyze student responses and identify key concepts related to the subject matter.

### **1.4.1 Text Processing Techniques**

Text processing techniques are used to convert raw textual data into a structured format that can be analyzed computationally. These techniques include tokenization, normalization, and keyword extraction. By applying these processes, the system can identify important terms within student responses and compare them with the expected answer patterns.

### **1.4.2 Semantic Analysis**

Semantic analysis focuses on understanding the meaning of words and sentences in context. Instead of simply matching keywords, semantic analysis helps the system interpret whether the student's answer reflects the correct concept. This improves the accuracy of automated evaluation systems and ensures that answers with similar meanings are recognized correctly.

### **1.4.3 Response Evaluation and Scoring**

Response evaluation involves analyzing student answers and assigning scores based on their correctness, completeness, and relevance. In intelligent evaluation systems, scoring algorithms compare the student response with reference answers or concept models. The system then generates evaluation results and feedback, helping students understand their strengths and areas for improvement.

Additionally, the response evaluation process can incorporate various analytical techniques to ensure fair and consistent scoring. The system may analyze factors such as the presence of important keywords, the logical structure of the response, and the semantic similarity between the student's answer and the expected answer. By using these evaluation criteria, the system can determine the depth of the student's understanding and provide an appropriate score. Furthermore, automated scoring helps maintain uniformity in evaluation, reduces the possibility of human bias, and allows students to receive timely feedback that can support their academic improvement.

## 1.5 SUMMARY

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) belongs to an interdisciplinary domain that integrates multiple advanced technological fields such as Artificial Intelligence (AI), Natural Language Processing (NLP), Educational Technology, and Language Analytics. These domains collectively contribute to the development of intelligent systems capable of understanding human communication, analyzing responses, and providing automated evaluation in academic environments. The combination of these technologies allows the system to simulate certain aspects of human examiners while maintaining consistency and efficiency in the evaluation process.

Artificial Intelligence forms the core foundation of this domain by enabling machines to perform tasks that typically require human intelligence, including reasoning, learning from data, and decision making. In educational applications, AI helps in automating various processes such as grading, feedback generation, and performance analysis. Within the context of viva-voce examinations, AI techniques can analyze student responses and determine whether the answer reflects the expected conceptual understanding. This reduces dependency on manual evaluation and ensures that the assessment process becomes more systematic and scalable.

Natural Language Processing plays a crucial role in enabling the system to understand and analyze human language. Since viva-voce examinations involve verbal or textual responses from students, NLP techniques are used to interpret these responses and convert them into structured data that can be analyzed computationally. Through processes such as tokenization, stop-word removal, syntactic analysis, and semantic similarity measurement, the system can evaluate the meaning and relevance of student answers. Overall, the domain of the AI-Driven Intelligent Viva-Voce Evaluation System represents a convergence of artificial intelligence, computational linguistics, and educational technology. This interdisciplinary approach enables the creation of innovative academic tools that transform traditional assessment methods into more efficient, data-driven, and intelligent evaluation processes.

## **CHAPTER 2**

### **LITERATURE SURVEY**

#### **2.1 AI-BASED AUTOMATED VIVA ASSESSMENT USING NLP – JAMES CARTER, ANITA SHARMA, AND LI WEI (2025)**

This research paper investigates the application of Artificial Intelligence (AI) and Natural Language Processing (NLP) technologies for automating the evaluation of viva-voce examinations in higher education. Traditional viva examinations are widely used to assess a student's conceptual understanding, communication ability, and analytical thinking. However, such assessments are often influenced by subjective judgment, examiner bias, and time limitations when evaluating a large number of students. These limitations can lead to inconsistencies in grading and may affect the fairness of the evaluation process. To address these challenges, the authors propose an AI-based automated viva assessment system that can analyze student responses using advanced language processing techniques and generate evaluation scores automatically.

The proposed system is designed to simulate a structured viva examination environment in which students respond to questions either through text or speech. The system then processes these responses using Natural Language Processing algorithms to determine the relevance and correctness of the answer. By automating the evaluation process, the system aims to improve consistency, transparency, and efficiency in academic assessments. The authors emphasize that AI-driven assessment systems can help institutions manage large-scale examinations more effectively while ensuring that students receive fair and unbiased evaluation.

A key component of the proposed framework is the use of transformer-based NLP models, which have shown significant improvements in language understanding tasks. These models are capable of capturing contextual relationships between words and sentences, enabling the system to interpret the meaning of student responses more accurately. Unlike traditional keyword-based grading systems, transformer models can analyze the semantic structure of answers and recognize correct explanations even when students use different vocabulary or sentence structures.

## **2.2 INTELLIGENT ORAL EXAMINATION SYSTEM USING MACHINE LEARNING – DAVID KIM AND RACHEL BROWN (2024)**

This research focuses on the development of an intelligent oral examination system powered by machine learning algorithms. The authors address the challenges faced by educational institutions in conducting and evaluating oral examinations for large groups of students. Traditional viva-voce examinations require significant time and effort from instructors, as each student must be evaluated individually through verbal interaction. Additionally, the evaluation process may sometimes vary depending on the examiner's judgment, which can result in inconsistencies in grading. To overcome these limitations, the authors propose an automated oral examination system that uses machine learning techniques to analyze and evaluate student responses efficiently.

The proposed system is designed to simulate a digital oral examination environment where students answer questions through speech or text. Student responses are captured and processed using speech recognition technology, which converts spoken answers into textual form. Once the responses are converted into text, the system applies Natural Language Processing techniques to analyze the structure and meaning of the responses. This integration of speech recognition and language processing enables the system to handle real-time oral responses in a manner similar to human examiners.

A key component of the proposed framework is the use of machine learning classification models trained on previously evaluated student responses. These models learn patterns associated with correct and incorrect answers by analyzing datasets containing sample responses and corresponding evaluation scores. Through this training process, the system becomes capable of predicting the correctness of new student responses. Machine learning algorithms such as decision trees, support vector machines, and neural networks are utilized to classify responses based on their similarity to expected answers.

The evaluation process begins with text preprocessing, which involves cleaning and structuring the response data. This step includes removing unnecessary words, identifying key terms, and extracting meaningful features from the student's answer.

### **2.3 NLP-BASED STUDENT RESPONSE ANALYSIS FOR AUTOMATED ASSESSMENT – EMILY JOHNSON, ARUN GUPTA, AND YUKI TANAKA (2023)**

This research paper investigates the use of Natural Language Processing techniques for analyzing and evaluating student responses in educational assessments. With the increasing adoption of digital learning platforms, educational institutions are seeking efficient methods for evaluating student performance automatically. Traditional evaluation systems often rely on manual grading, which can be time-consuming and may lead to inconsistencies when dealing with large numbers of students. To address this challenge, the authors propose an NLP-based response analysis framework designed to evaluate descriptive answers automatically.

The study highlights that many automated grading systems rely primarily on keyword matching techniques, which may not accurately capture the meaning of student responses. Students may express the same concept using different words or sentence structures, making simple keyword matching insufficient for reliable evaluation. To overcome this limitation, the authors propose an evaluation framework that focuses on semantic similarity analysis, allowing the system to understand the meaning behind the student's response rather than simply matching specific keywords.

The proposed framework uses word embeddings and contextual language models to represent words and sentences in a numerical format that captures their semantic relationships. Word embedding techniques map words with similar meanings to nearby points in a vector space, enabling the system to identify conceptually related terms. By applying these techniques, the system can evaluate responses even when students use synonyms or alternative phrasing to explain a concept.

The response analysis process begins with text preprocessing, which involves breaking down the student's response into smaller components for analysis. Techniques such as tokenization, stop-word removal, and stemming are applied to extract meaningful words from the text. These processed words are then converted into vector representations using word embedding models.

## **2.4 AUTOMATED ORAL EXAMINATION PLATFORM USING AI – MICHAEL THOMPSON AND PRIYA NARAYANAN (2022)**

This study presents an Artificial Intelligence-based oral examination platform designed to evaluate student knowledge through automated questioning and response analysis. Oral examinations have long been considered an effective method for assessing a student's understanding of concepts, communication skills, and critical thinking abilities. However, conducting oral examinations manually can be time-consuming, especially when evaluating large numbers of students. The authors propose an automated oral examination system that uses artificial intelligence technologies to streamline the evaluation process and improve the efficiency of academic assessments.

The proposed system is designed to function as an interactive examination platform where students respond to questions generated by the system. The platform supports both text-based and speech-based responses, allowing students to answer questions in a natural and flexible manner. The system then processes these responses using Natural Language Processing algorithms to determine their relevance and correctness with respect to the given question.

One of the key features of the system is its automated questioning mechanism, which allows instructors to create question banks covering various topics within a subject. During the examination, the system selects questions dynamically and presents them to students. This feature helps ensure that each student receives a unique set of questions, reducing the possibility of repetition or memorization of answers.

After receiving the student's response, the system performs several stages of language processing to evaluate the answer. These stages include text preprocessing, keyword extraction, and semantic similarity analysis. By comparing the student's response with predefined reference answers, the system determines whether the response contains the essential concepts required to answer the question correctly.

The authors emphasize that automated evaluation systems can significantly improve the consistency and fairness of academic assessments. Since the evaluation process is based on predefined algorithms and language analysis techniques.

## **2.5 NATURAL LANGUAGE UNDERSTANDING FOR EDUCATIONAL ASSESSMENT – ROBERT MILLER AND SNEHA PATEL (2021)**

This research investigates the role of Natural Language Understanding (NLU) in improving automated evaluation systems used in educational environments. The authors highlight that traditional automated grading systems often rely on simple keyword matching techniques, which may not accurately capture the meaning of student responses. Since students can express the same idea using different words, sentence structures, or explanations, keyword-based systems may incorrectly evaluate correct answers as incorrect. To address this limitation, the authors propose an advanced response evaluation framework based on Natural Language Understanding techniques that focus on interpreting the semantic meaning and contextual relationships present in student responses.

The proposed framework integrates multiple levels of language analysis, including syntactic processing and semantic understanding, to analyze student responses more effectively. The first stage of the system involves text preprocessing, where the student's response is cleaned and structured by removing unnecessary characters and identifying meaningful words. Techniques such as tokenization, stop-word removal, and part-of-speech tagging are applied to convert the raw response into a format suitable for computational analysis.

Following preprocessing, the system performs dependency parsing, which analyzes grammatical relationships between words in a sentence. This step helps the system understand how different words relate to each other and how the overall meaning of the sentence is constructed. By identifying subject–verb–object relationships and other grammatical dependencies, the system can better interpret the logical structure of the student's answer.

Another important component of the system is semantic similarity analysis, which measures how closely the student's response matches the expected answer in terms of meaning. Instead of focusing solely on exact word matches, the system evaluates whether the key concepts and ideas in the student's answer correspond to the correct explanation.

## **2.6 MACHINE LEARNING APPROACHES FOR AUTOMATED STUDENT EVALUATION – THOMAS LEE AND SARAH WILLIAMS (2020)**

This research paper explores the application of machine learning techniques for automated evaluation of student responses in academic assessments. With the growing number of students enrolled in online and traditional educational programs, instructors face increasing challenges in grading assignments and examination responses efficiently. Manual evaluation of descriptive answers requires significant time and effort, and it may also introduce inconsistencies due to subjective judgment. To address these challenges, the authors propose the use of machine learning models that can automatically analyze student responses and assign evaluation scores based on learned patterns.

The proposed system is designed to analyze textual responses provided by students during examinations or assignments. The system begins with text preprocessing, where student responses are cleaned and structured to remove unnecessary characters and irrelevant words. Techniques such as tokenization, stemming, and stop-word removal are applied to extract meaningful information from the text. This preprocessing step helps convert unstructured textual data into a structured representation that can be used by machine learning algorithms.

Once the text is preprocessed, the system extracts relevant features from the student responses. These features may include keyword frequency, semantic similarity scores, sentence structure patterns, and contextual relationships between words. The extracted features are then used as input for various machine learning algorithms that classify responses based on their correctness.

The authors compare several machine learning algorithms, including Support Vector Machines (SVM), Decision Trees, and Artificial Neural Networks (ANN). Each algorithm is trained using datasets containing previously evaluated student responses and corresponding grading labels. Through the training process, the models learn patterns that distinguish correct answers from incorrect ones.

## **2.7 AUTOMATED GRADING SYSTEM USING NLP TECHNIQUES – DANIEL MARTINEZ AND KAVYA REDDY (2019)**

This paper discusses the design and development of an automated grading system based on Natural Language Processing techniques. The authors highlight that evaluating descriptive answers in examinations is one of the most challenging tasks in educational assessment. Unlike multiple-choice questions, descriptive answers require detailed analysis of the student's explanation, reasoning, and conceptual understanding. Manual evaluation of such responses can be time-consuming and may lead to inconsistencies when different instructors interpret answers differently. To address these challenges, the authors propose an automated grading system that uses NLP algorithms to analyze student responses and determine their similarity to reference answers.

The proposed system begins by performing text preprocessing on the student responses. This process includes tokenization, which breaks the text into individual words or tokens. Stop-word removal is then applied to eliminate common words such which do not contribute significantly to the meaning of the response. Stemming techniques are used to reduce words to their root forms, allowing the system to recognize related words that share the same base meaning.

After preprocessing, the system performs keyword extraction to identify the important terms present in the student's answer. These keywords are compared with the key concepts present in the reference answer provided by instructors. The system then calculates semantic similarity scores using NLP techniques to determine how closely the student's response matches the expected explanation.

The evaluation algorithm considers multiple factors when assigning scores, including the presence of essential keywords, the completeness of the explanation, and the overall similarity between the student's response and the reference answer. By combining these factors, the system generates an evaluation score that reflects the quality of the student's response.

## **CHAPTER 3**

### **EXISTING SYSTEM**

In most educational institutions, the viva-voce examination is conducted manually by faculty members or examiners. During this process, students appear individually before an examiner or a panel of examiners and answer questions related to their academic subjects or project work. The examiner asks questions verbally, and the student provides responses based on their knowledge and understanding of the topic. The examiner then evaluates the student's performance based on factors such as correctness of the answer, clarity of explanation, confidence, and communication skills.

The traditional viva-voce system plays an important role in assessing a student's conceptual understanding, analytical ability, and communication skills. Unlike written examinations, oral examinations allow examiners to interact directly with students and evaluate their reasoning and problem-solving abilities in real time. However, the entire evaluation process depends heavily on the examiner's judgment and experience.

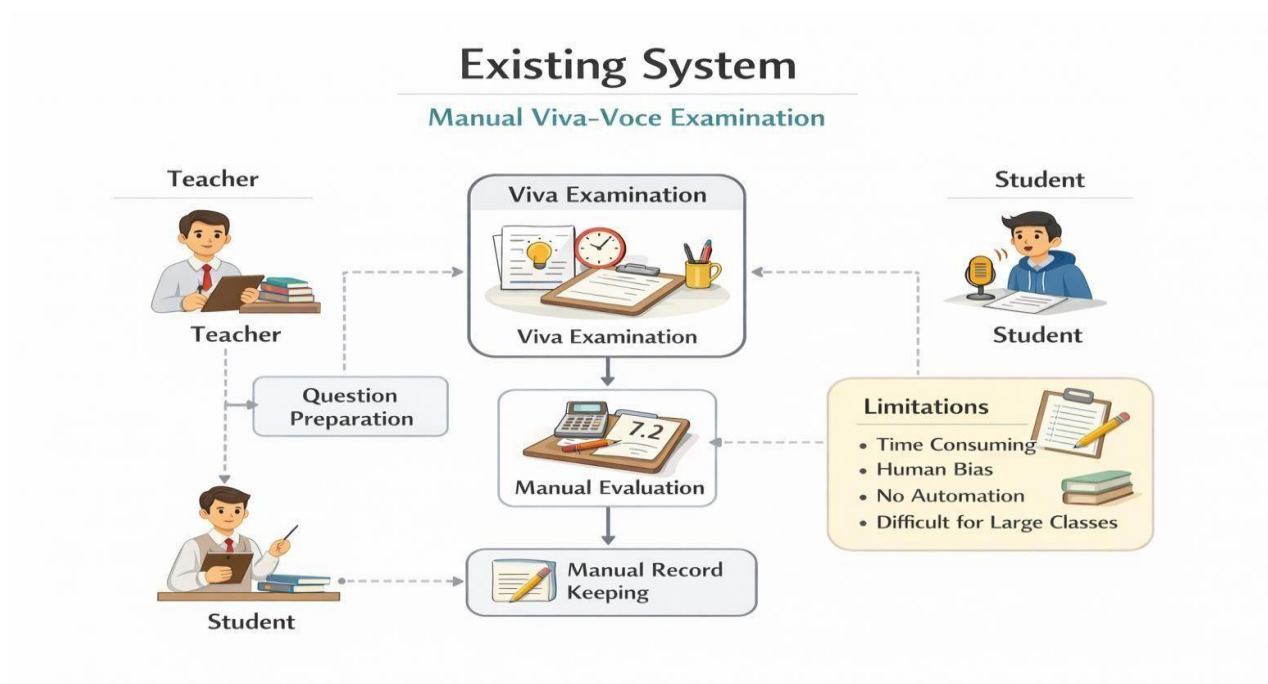
In the existing system, the evaluation process is completely manual. The examiner listens to the student's response, interprets the explanation, and assigns marks based on personal evaluation criteria. Since the assessment is conducted by different examiners, the evaluation standards may vary from one examiner to another. As a result, there can be differences in grading even when students provide similar answers.

Another limitation of the traditional system is the time required to conduct viva examinations. When the number of students is large, examiners must spend significant time interviewing each student individually. This process can be time-consuming and may create scheduling difficulties in educational institutions. Faculty members must dedicate several hours or even days to complete the viva evaluation process.

The existing system also lacks proper digital documentation and analysis of student responses. Most viva evaluations are recorded only as final marks without maintaining detailed records of the responses provided by students.

Furthermore, the traditional viva process may sometimes introduce subjective bias, where the evaluation may be influenced by the examiner's perception, mood, or interpretation of the answer. Students who express the same concept in different ways may receive different scores depending on how the examiner interprets their response.

Although the traditional viva-voce examination system has been widely used for many years, the growing number of students and the increasing adoption of digital education technologies highlight the need for more efficient and consistent evaluation methods. These limitations in the existing system create the need for intelligent solutions that can support automated response analysis and evaluation.



**Figure.3.1 Existing System Diagram**

## **CHAPTER 4**

### **PROBLEMS IDENTIFIED**

The existing viva-voce examination system faces several challenges that affect the efficiency and reliability of the evaluation process. One of the major problems is the subjectivity in evaluation. Since different examiners may have different evaluation criteria, the grading process may not always be consistent. This can lead to variations in marks even when students provide similar answers.

Another problem is the time-consuming nature of manual viva examinations. When the number of students is high, conducting individual viva sessions becomes a lengthy process. Examiners must allocate significant time to ask questions, listen to responses, and record marks for each student. This can delay the completion of the examination process.

The system also lacks automated analysis of student responses. In the current approach, responses are evaluated only through human interpretation, and there is no technological support to analyze the content of the student's answer. As a result, valuable insights about student understanding, knowledge gaps, and performance trends are not captured.

Another limitation is the lack of proper feedback mechanisms. In many cases, students receive only their final marks without detailed feedback explaining which parts of their answer were correct or incorrect. This makes it difficult for students to identify their weaknesses and improve their understanding of the subject.

The manual evaluation system also creates difficulty in handling large-scale assessments. Educational institutions with large student populations may struggle to conduct viva examinations efficiently within limited timeframes. Managing schedules, recording results, and ensuring consistent evaluation standards can become challenging for faculty members.

## CHAPTER 5

### PROPOSED SYSTEM

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) is proposed to overcome the limitations of traditional viva-voce examination systems. This system integrates Artificial Intelligence, Natural Language Processing, and machine learning technologies to automate the process of conducting and evaluating oral examinations.

In the proposed system, students interact with an intelligent digital platform that presents viva questions related to the subject or project topic. Students can provide responses either in text form or through speech input. If the response is given through speech, speech recognition technology converts the spoken response into text so that it can be processed by the system.

Once the response is received, the system applies Natural Language Processing techniques to analyze the content of the answer. The response is first processed using text preprocessing techniques such as tokenization, stop-word removal, and stemming. These processes help extract meaningful words and concepts from the student's answer.

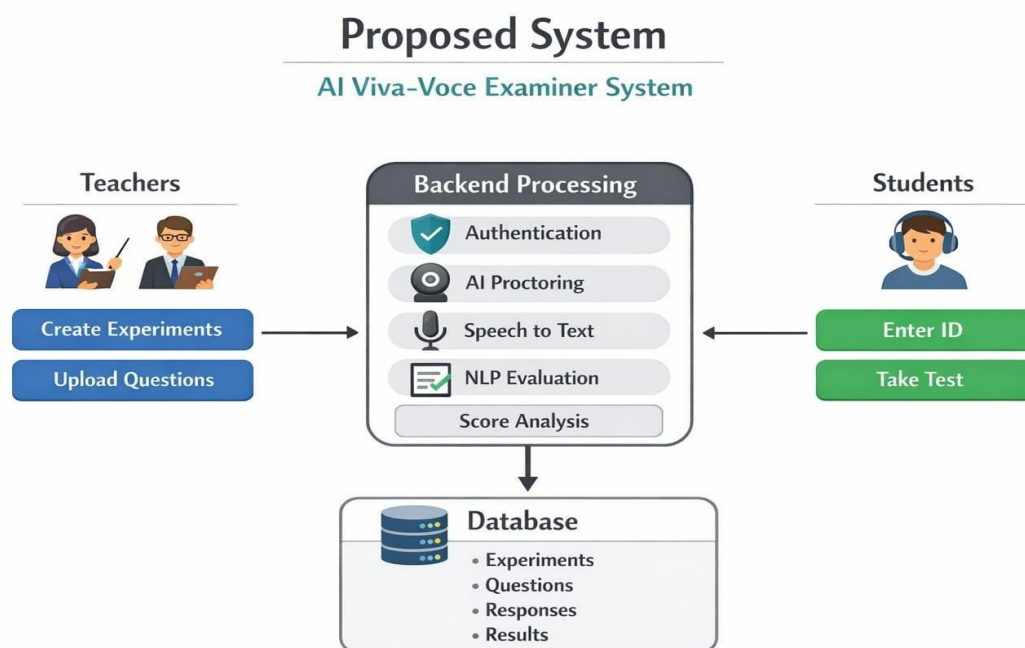
After preprocessing, the system performs semantic similarity analysis to compare the student's response with reference answers stored in the system database. Instead of relying solely on keyword matching, the system analyzes the meaning and context of the response to determine whether the student has explained the concept correctly.

Machine learning algorithms are used to classify the response and generate an evaluation score based on the correctness, relevance, and completeness of the answer. The system can also provide instant feedback to students, highlighting which parts of the answer were correct and which areas require improvement.

Another advantage of the proposed system is its ability to maintain digital records of student responses and evaluation results. These records can be used for performance analysis, academic research, and continuous improvement of the evaluation system.

The proposed AI-based system ensures consistency, transparency, and efficiency in the evaluation process. Since the evaluation criteria are defined through algorithms and trained models, all students are assessed using the same standards. This reduces examiner bias and improves the fairness of the assessment process.

By automating the viva-voce evaluation process, the system also helps reduce the workload of instructors and allows institutions to conduct assessments more efficiently, even with a large number of students.



**Figure. 5.1 Proposed System Diagram**

## **CHAPTER 6**

### **SYSTEM REQUIREMENTS**

#### **6.1 HARDWARE REQUIREMENTS**

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher
- Input Devices: Keyboard and Microphone
- Output Devices: Monitor or Display
- Internet Connection: Required for online access and AI model integration

#### **6.2 SOFTWARE REQUIREMENTS**

- Operating System: Windows / Linux / macOS
- Programming Language: Python / Java
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Framework: Flask / Django / Node.js
- Database: MySQL / MongoDB
- Libraries for AI and NLP:
  - NLTK
  - spaCy
  - TensorFlow / PyTorch
- Speech Recognition Tools: Google Speech API or similar tools

## **CHAPTER 7**

### **SYSTEM IMPLEMENTATIONS**

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) is implemented as an intelligent software platform that automates the process of conducting viva examinations and evaluating student responses. The system integrates several technologies such as Artificial Intelligence, Natural Language Processing, speech recognition, and database management to analyze student answers and generate evaluation results.

The implementation of the system is divided into several modules, each responsible for performing a specific function in the overall evaluation process. These modules work together to ensure smooth interaction between students and the system while maintaining accurate and efficient analysis of responses. The modular architecture also makes the system easier to maintain, update, and expand with additional features in the future.

The system begins by allowing students to log in and access the viva examination interface. The system then presents questions to the student, records their responses, and processes the answers using NLP algorithms. After analyzing the responses, the system evaluates the answer and generates scores and feedback. All responses and evaluation results are stored in a database for future reference and analysis.

#### **7.1 LIST OF MODULES**

The system is divided into the following modules:

1. User Authentication Module
2. Question Management Module
3. Student Response Processing Module
4. NLP-Based Answer Evaluation Module
5. Result Generation and Feedback Module.

## **7.2 MODULE DESCRIPTION**

### **7.2.1 User Authentication Module**

The User Authentication Module is responsible for managing the login and access control of the system. This module ensures that only authorized users such as students and administrators can access the platform. Students must log in using their credentials before starting the viva examination session.

The authentication system verifies the user's identity by validating login details stored in the system database. Once authenticated, the student is granted access to the viva interface where questions will be presented. The module also helps maintain security by preventing unauthorized access to examination data.

Additionally, administrators can use this module to manage user accounts, update login credentials, and control system access. By implementing authentication mechanisms, the system ensures data privacy and secure management of examination information.

### **7.2.2 Question Management Module**

The Question Management Module is responsible for storing and managing viva questions used in the examination process. This module allows instructors or administrators to add, modify, and organize questions related to different subjects or topics.

Questions are stored in a structured database along with reference answers that will later be used for response evaluation. The system can randomly select questions from the database to present them to students during the viva session. This helps maintain fairness and prevents repetition of questions across different students.

### **7.2.3 Student Response Processing Module**

The Student Response Processing Module is responsible for capturing and preparing student responses for analysis. Students can provide answers either in text form or through voice input. If the response is spoken, speech recognition technology is used to convert the speech into text format.

After receiving the response, the system performs initial preprocessing tasks to clean and structure the data. This includes removing unnecessary characters, converting text into lowercase, and preparing the response for further analysis. These preprocessing steps help improve the accuracy of the evaluation process.

### **7.2.4 NLP-Based Answer Evaluation Module**

The NLP-Based Answer Evaluation Module is the core component of the intelligent viva evaluation system. This module analyzes student responses using Natural Language Processing techniques to determine whether the answer is correct and relevant to the question.

The system first performs text preprocessing techniques such as tokenization, stop-word removal, and stemming to extract meaningful information from the student's answer. After preprocessing, the system compares the response with predefined reference answers using semantic similarity analysis.

Machine learning algorithms may also be used to identify patterns in student responses and classify answers based on their correctness. By analyzing both keywords and contextual meaning, the system can evaluate answers even when students use different wording to explain the same concept.

### 7.2.5 Result Generation and Feedback Module

The Result Generation and Feedback Module is responsible for generating the final evaluation results after analyzing student responses. Once the NLP evaluation process is completed, the system assigns scores based on the correctness, relevance, and completeness of the student's answer. The scoring mechanism compares the processed response with predefined reference answers and determines how closely the student's explanation matches the expected concept.

The module also generates feedback that helps students understand their performance. Feedback may include suggestions for improvement, explanations about missing concepts, or indications of partially correct responses. This feature allows students to identify areas where their understanding may be incomplete and encourages them to review the topic for better learning outcomes.

All evaluation results are stored in the system database for future reference. Administrators and instructors can access these records to analyze student performance, track academic progress, and generate performance reports. The stored data can also be used for statistical analysis, helping educational institutions understand overall learning trends and identify common difficulties faced by students.

In addition, the module supports the generation of performance summaries and evaluation reports. These reports may include details such as the number of questions attempted, accuracy of responses, average scores, and feedback comments. Such reports can be useful for both instructors and students in monitoring academic performance over time.

The module may also include a dashboard interface where students can view their evaluation results immediately after completing the viva session. This real-time feedback system improves transparency in the evaluation process and allows students to quickly understand their strengths and weaknesses.

## **CHAPTER 8**

### **SYSTEM TESTING**

#### **8.1 UNIT TESTING**

Unit testing focuses on testing individual components or modules of the system separately to ensure that each part functions correctly. In the AI-Driven Viva-Voce Evaluation System, unit testing is performed on modules such as the user authentication module, question management module, response processing module, NLP evaluation module, and result generation module.

Each module is tested independently by providing sample inputs and verifying the expected outputs. For example, the login module is tested by entering valid and invalid credentials to ensure that authentication works properly. Similarly, the response processing module is tested by providing different types of student answers to verify that the system correctly captures and processes the input.

Unit testing helps detect errors in individual modules at an early stage of development. By identifying issues during this stage, developers can correct them before integrating the modules into the complete system.

#### **8.2 INTEGRATION TESTING**

Integration testing is performed after unit testing to verify that different modules of the system work correctly when combined together. In this stage, interactions between modules are tested to ensure smooth data flow and communication.

For example, when a student logs into the system, the authentication module must communicate with the question management module to display questions. After the student submits an answer, the response processing module must transfer the processed data to the NLP evaluation module for analysis. Finally, the result generation module must display the evaluation results to the user.

### **8.3 SYSTEM TESTING**

System testing is performed on the complete system to verify that it meets the specified functional requirements. In this stage, the entire AI-based viva evaluation platform is tested as a whole.

Different scenarios are tested to ensure that the system behaves as expected. For example, students are asked to log in, answer questions, and receive evaluation results. The system is tested to ensure that it correctly processes responses, analyzes answers using NLP techniques, and generates appropriate scores.

System testing also verifies that the user interface functions properly and that the system can handle multiple users without affecting performance. This stage ensures that the system is ready for real-world usage in educational institutions.

### **8.4 PERFORMANCE TESTING**

Performance testing is conducted to evaluate how efficiently the system performs under different conditions. The purpose of this testing is to ensure that the system can handle a large number of student responses without delays or system failures.

During performance testing, the system is tested with multiple users submitting answers simultaneously. The response time of the system, processing speed of NLP algorithms, and database performance are monitored to ensure that the system operates efficiently.

The testing results confirm that the system is capable of processing responses quickly and generating evaluation results within a short period of time. This ensures that the platform can support large-scale academic assessments.

## **8.5 SECURITY TESTING**

Security testing is conducted to ensure that the system protects sensitive data and prevents unauthorized access. Since the system handles student information, login credentials, and examination results, it is important to maintain strong security measures.

During this testing phase, the authentication system is tested to verify that only authorized users can access the platform. Password validation, session management, and database protection mechanisms are evaluated to ensure that user data remains secure.

Security testing also ensures that the system is protected against common threats such as unauthorized login attempts, data manipulation, and system vulnerabilities. This helps maintain the integrity and confidentiality of academic information.

## **8.6 USABILITY TESTING**

Usability testing evaluates how easy and convenient the system is for users to interact with. The goal of this testing is to ensure that students and instructors can use the system without difficulty.

During usability testing, users interact with the system interface to perform tasks such as logging in, answering questions, and viewing results. Feedback is collected regarding the clarity of instructions, ease of navigation, and overall user experience.

The testing results indicate that the system interface is user-friendly and intuitive. Students can easily access the examination platform, provide responses, and view evaluation results without confusion. The simple interface design improves the overall usability of the system and ensures a positive user experience.

## **CHAPTER 9**

### **RESULTS AND DISCUSSION**

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing (NLP) was developed to automate the process of conducting and evaluating viva examinations. The system was tested using different types of questions and student responses to evaluate its effectiveness in analyzing answers and generating accurate results.

During the testing phase, students were asked to log into the system and answer viva questions related to specific academic topics. The system successfully captured student responses and processed them using Natural Language Processing techniques. The responses were analyzed through several stages including text preprocessing, keyword extraction, semantic similarity analysis, and evaluation scoring.

The results obtained from the system indicate that the proposed platform is capable of identifying relevant concepts in student answers and assigning evaluation scores based on the correctness and completeness of the response. The system demonstrated the ability to recognize answers even when students used different words or sentence structures to express the same concept. This shows the effectiveness of NLP techniques in understanding the meaning and context of textual responses.

The system also generated instant evaluation results and feedback for students. The feedback helped students understand which parts of their answers were correct and which areas required improvement. This feature improves the learning experience by allowing students to immediately review their performance and strengthen their knowledge.

Another important outcome observed during testing was the reduction in manual workload for instructors. Since the system automatically evaluates student responses, instructors are not required to conduct lengthy viva sessions for each student.

## **CHAPTER 10**

### **CONCLUSION AND FUTURE WORK**

#### **10.1 CONCLUSION**

The AI-Driven Intelligent Viva-Voce Evaluation System using Natural Language Processing was developed to address the limitations of traditional viva examination methods. The system integrates artificial intelligence and language processing technologies to automate the evaluation of student responses and provide accurate assessment results.

Through the implementation of NLP techniques, the system is capable of analyzing student answers, identifying key concepts, and comparing responses with reference answers to determine their correctness. This allows the system to evaluate descriptive responses effectively without relying solely on keyword matching.

The system provides several advantages compared to traditional viva examination methods. It improves evaluation consistency by applying the same scoring criteria to all students, thereby reducing the possibility of human bias. It also reduces the time required for conducting viva examinations and helps instructors manage assessments more efficiently.

Another important contribution of the system is the ability to generate instant feedback for students. This feature enhances the learning process by allowing students to understand their mistakes and improve their conceptual knowledge. Additionally, the system maintains digital records of responses and results, which can be used for performance analysis and academic monitoring.

Overall, the proposed system demonstrates how Artificial Intelligence and Natural Language Processing technologies can be effectively used to modernize the viva examination process. The system improves efficiency, transparency, and reliability in academic evaluation and represents an important step toward intelligent educational assessment systems.

## 10.2 FUTURE ENHANCEMENTS

Although the proposed system performs effectively in analyzing textual responses, several improvements can be made in future versions to enhance its capabilities.

One possible enhancement is the integration of advanced speech recognition technology to support real-time voice-based viva examinations. This would allow students to answer questions verbally, making the system more similar to traditional oral examinations.

Another improvement could involve the use of deep learning models and transformer-based NLP architectures to improve the accuracy of response evaluation. These advanced models can better understand context, complex sentence structures, and conceptual explanations provided by students.

Future versions of the system may also include adaptive questioning mechanisms, where the system dynamically adjusts the difficulty level of questions based on the student's previous answers. This would create a more personalized assessment experience for students.

Additionally, the system could be integrated with learning management systems (LMS) used by educational institutions. Such integration would allow instructors to manage examinations, track student performance, and analyze learning outcomes more effectively.

Another possible enhancement is the development of advanced analytics dashboards that provide detailed insights into student performance trends, common errors, and subject-wise evaluation statistics. These insights can help instructors improve teaching strategies and support student learning.

By incorporating these enhancements, the AI-driven viva evaluation system can evolve into a more intelligent and comprehensive academic assessment platform capable of supporting modern digital education environments.

## APPENDIX A

### SOURCE CODE

```
// src/pages/index.tsx

import { useEffect, useState } from "react";
import { supabase } from "@lib/supabaseClient";
import { useRouter } from "next/router";

type SoftUser = { email: string; role: "student" | "teacher" };
const SOFT_KEY = "ai_viva_soft_user";

export default function Home() {
  const [email, setEmail] = useState("");
  const [role, setRole] = useState<"student" | "teacher">("student");
  const [msg, setMsg] = useState("");
  const [sessionUser, setSessionUser] = useState<any | null>(null);
  const [loading, setLoading] = useState(false);
  const router = useRouter();

  // Load real Supabase session and also preload any soft-login stored user.
  useEffect(() => {
    let mounted = true;
    (async () => {
      try {
        const r = await supabase.auth.getSession();
        const su = r.data.session?.user ?? null;
        if (!mounted) return;
        if (su) {
          setSessionUser(su);
          setEmail(su.email || "");
        }
      }
    })();
  });
}
```

```

    } else {
      // try soft-login
      if (typeof window !== "undefined") {
        const raw = localStorage.getItem(SOFT_KEY);
        if (raw) {
          try {
            const u: SoftUser = JSON.parse(raw);
            setSessionUser({ email: u.email, soft: true, role: u.role });
            setEmail(u.email);
            setRole(u.role);
          } catch {
            localStorage.removeItem(SOFT_KEY);
          }
        }
      }
    } catch (err) {
      console.error("session load error", err);
    }
  })();

  const { data: sub } = supabase.auth.onAuthStateChange((_event, session) => {
    setSessionUser(session?.user ?? null);
    if (session?.user?.email) setEmail(session.user.email);
  });

  return () => {
    mounted = false;
    try {
      sub.subscription.unsubscribe();
    }
  }
}

```

```

    } catch {}
  };
}, []);

// helper: validate simple email
function isValidEmail(e?: string) {
  if (!e) return false;
  return /^[S+@\S+\.\S+]/.test(e.trim());
}

// soft login (local). For teacher role we auto-insert into approved_emails role='teacher'
async function softLogin() {
  setMsg("");
  if (!isValidEmail(email)) {
    setMsg("Enter a valid institutional email.");
    return;
  }
  const normalized = email.trim().toLowerCase();
  setLoading(true);
  try {
    if (role === "teacher") {
      // Check existing approved_emails entry
      const { data: existing, error: lookupErr } = await supabase
        .from("approved_emails")
        .select("role")
        .eq("email", normalized)
        .maybeSingle();

      if (lookupErr) {
        // ignore DB missing table errors in dev, but show warning

```

```

    console.warn("approved_emails lookup error", lookupErr);
  }

  if (existing && existing.role && existing.role !== "teacher") {
    setMsg("This email is already registered as a student. Duplicate roles are not
allowed.");
    setLoading(false);
    return;
  }

  if (!existing) {
    // try to insert teacher record (if unique constraint fails, ignore)
    const { error: insertErr } = await supabase.from("approved_emails").insert([ {
email: normalized, role: "teacher" } ]);
    if (insertErr) {
      // If another process inserted concurrently, that's okay; if it's a real error, surface
it.
      if (!/unique|duplicate/i.test(insertErr.message || "")) {
        console.warn("approved_emails insert error", insertErr);
      }
    }
  }
} else {
  // student soft-login: do not auto-register student in approved_emails table.
  // Teachers control which students can attend via experiment_approved_emails.
}

// store local soft-login
const soft: SoftUser = { email: normalized, role };
localStorage.setItem(SOFT_KEY, JSON.stringify(soft));
setSessionUser({ email: normalized, soft: true, role });

```

```

    setMsg(`Logged in locally as ${normalized} (${role}).`);
  } catch (err: any) {
    console.error("softLogin err", err);
    setMsg("Login failed: " + (err?.message || String(err)));
  } finally {
    setLoading(false);
  }
}

```

// clear soft-login and supabase session

```

async function signOut() {
  setMsg("");
  try {
    localStorage.removeItem(SOFT_KEY);
    // attempt to sign out supabase session if present (no harm if not)
    await supabase.auth.signOut();
    setSessionUser(null);
    setMsg("Signed out (local).");
  } catch (err: any) {
    console.warn(err);
    setMsg("Signed out locally.");
    setSessionUser(null);
  }
}

```

// Proceed to dashboard or student portal.

```

async function handleProceed() {
  setMsg("");
  if (!isValidEmail(email)) {
    setMsg("Enter a valid institutional email.");
  }
}

```

```

    return;
}

// Must have a soft or supabase session
if (!sessionUser) {
    setMsg("Please login using the Login button on this page first (local login).");
    return;
}

// ensure typed email matches logged-in/soft email
const currentEmail = (sessionUser.email || "").toLowerCase();
if (currentEmail !== email.trim().toLowerCase()) {
    setMsg(`You're currently logged in as ${currentEmail}. Change the email field or
    Sign out and login with ${email.trim()}.`);
    return;
}

if (role === "teacher") {
    // teacher route
    router.push("/teacher/dashboard");
    return;
}

// student route
router.push("/student/attend");
}

return (
    <div className="min-h-screen flex items-center justify-center bg-zinc-50 p-6">
        <div className="max-w-xl w-full bg-white p-8 rounded shadow card">
            <h1 className="text-2xl font-bold mb-3">AI Viva-Voce Examiner</h1>

```

```

<div className="mb-4">
  <label className="block text-sm font-medium">I am a</label>
  <select value={role} onChange={(e) => setRole(e.target.value as any)}
className="mt-1 w-full border p-2 rounded">
    <option value="student">Student</option>
    <option value="teacher">Teacher / Staff</option>
  </select>
</div>

```

```

<div className="mb-4">
  <label className="block text-sm font-medium">Your institutional
Email</label>
  <input
    value={email}
    onChange={(e) => setEmail(e.target.value)}
    className="mt-1 w-full border p-2 rounded"
    placeholder="your@college.edu"
    type="email"
  />
</div>

```

```

<div className="flex gap-2">
  <button onClick={softLogin} className="bg-blue-600 text-white px-4 py-2
rounded" disabled={loading}>
    Login (no magic link)
  </button>

  <button onClick={handleProceed} className="border px-4 py-2 rounded">
    Proceed to {role === "teacher" ? "Teacher Dashboard" : "Student Portal"}
  </button>

```

```

    {sessionUser ? (
      <button onClick={signOut} className="border px-4 py-2 rounded ml-auto">
        Sign out ({sessionUser.email})
      </button>
    ) : null}
  </div>

  <p className="mt-2 text-sm text-gray-600">{msg}</p>

  <hr className="my-4" />

  <div className="text-sm">
    <div className="font-medium mb-2">Notes:</div>
    <ul className="list-disc ml-5 mt-2 text-sm">
      <li>This page supports a lightweight local login so teachers and students can
sign in without magic links for normal app access.</li>
      <li>Teachers who login locally are auto-registered (created) in
<code>approved_emails</code> as <code>role='teacher'</code> so they can create
experiments immediately.</li>
      <li>Students still need to be added to an experiment's approved list
(<code>experiment_approved_emails</code>) by the teacher to attend that particular
test.</li>
      <li>Magic links are still sent when a teacher creates an experiment (optional).
They are useful if you want a student to open an authenticated session, but they are not
required for basic login.</li>
    </ul>
  </div>
</div>
);
}

```

## APPENDIX B

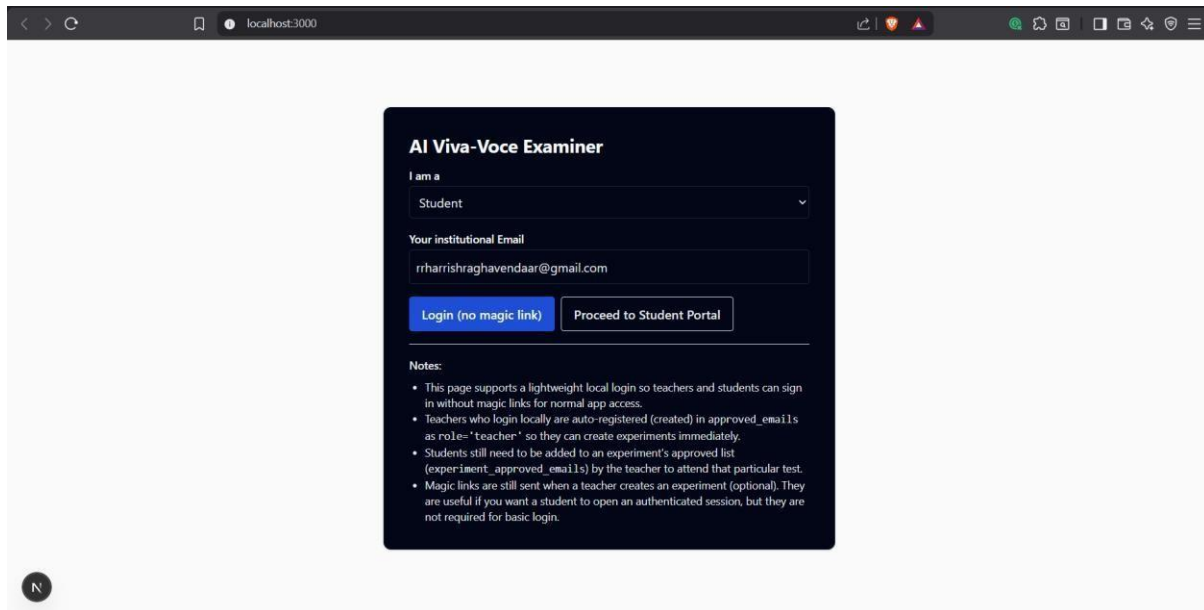


Figure. B.1. Login Page

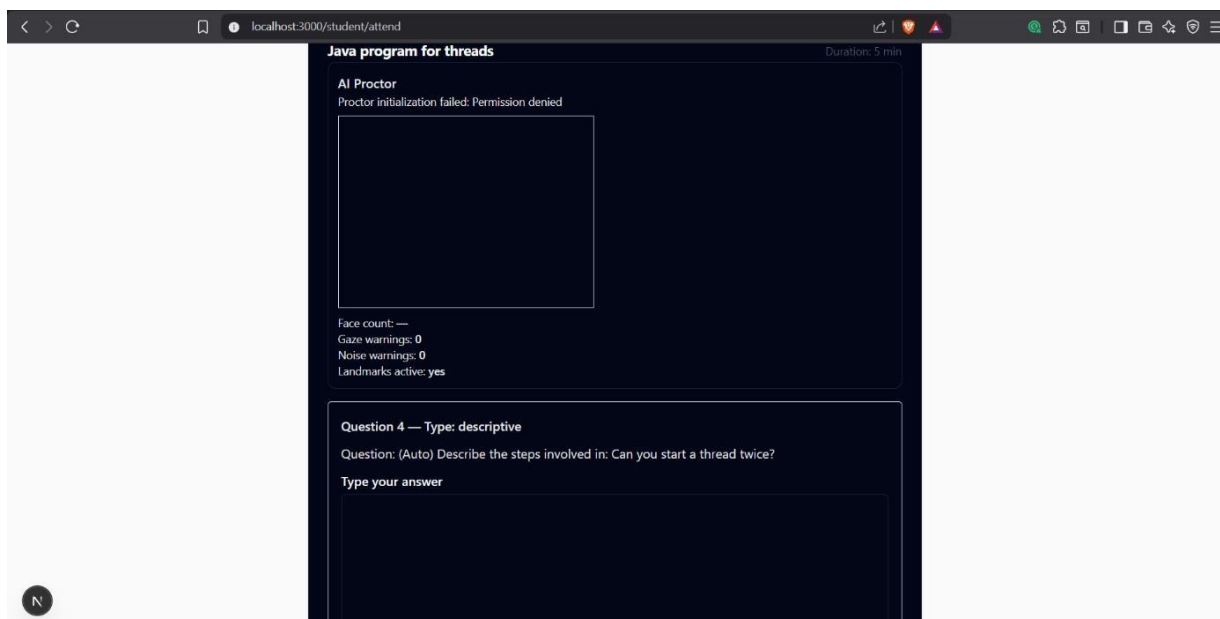


Figure. B.2. Insert Description

The screenshot shows a web browser window with the address bar displaying 'localhost:3000/teacher/dashboard'. The page title is 'Teacher Dashboard' and it indicates the user is signed in as 'rrhr100@gmail.com'. The main form for creating an experiment includes fields for 'Experiment Title', 'Experiment ID (students enter this)', 'Total Questions (optional)', 'Total Score (optional)', and 'Duration (minutes)'. Below these fields is a text input for 'Approved emails for this experiment (comma separated)' with the example 'student1@school.edu, student2@school.edu'. A green button labeled 'Create Experiment & Invite Students' is positioned below the email input. The 'Your Experiments' section lists four existing experiments: 'Java program for threads' (ID: 12, 10 questions, 20 total score, 5 min duration), 'Java program for recursion' (ID: 11, 10 questions, 20 total score), 'Java Program for Patterns' (ID: 10, 10 questions, 20 total score), and 'C Program for Recursion' (ID: 02, 10 questions, 20 total score). Each experiment entry has a link to 'Add / Auto-generate Questions' and a 'View Submissions' link. A small circular icon with the letter 'N' is visible in the bottom left corner of the dashboard area.

Teacher Dashboard

Signed in as: rrhr100@gmail.com

Experiment Title

Experiment ID (students enter this)

Total Questions (optional)

Total Score (optional)

Duration (minutes)

Approved emails for this experiment (comma separated)

student1@school.edu, student2@school.edu

Create Experiment & Invite Students

Your Experiments

Java program for threads

ID: 12 — Questions: 10 — Total Score: 20 — Duration: 5 min

Add / Auto-generate Questions View Submissions

Java program for recursion

ID: 11 — Questions: 10 — Total Score: 20 — Duration: —

Add / Auto-generate Questions View Submissions

Java Program for Patterns

ID: 10 — Questions: 10 — Total Score: 20 — Duration: —

Add / Auto-generate Questions View Submissions

C Program for Recursion

ID: 02 — Questions: — — Total Score: — — Duration: —

Add / Auto-generate Questions View Submissions

Figure. B.3. Create Experiment

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